L46 Project

Exploring Quantisation In FPGA-Accelerated Binarized Neural Networks

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1 Introduction

The rapid advancement of machine learning has driven an increasing demand for efficient hardware acceleration of neural networks. While general-purpose processors such as CPUs and GPUs have traditionally been used for this purpose, significant performance and energy efficiency gains have emerged from the use of specialised hardware accelerators. In particular, field-programmable gate arrays (FPGAs) have gained attention due to their ability to provide flexible, fine-grained parallelism and custom dataflow architectures tailored to specific workloads.

A key technique enabling these improvements is quantisation, which reduces the numerical precision of neural network parameters and activations. By representing values with fewer bits, ranging from reduced fixed-point precision to fully binarized representations, quantisation significantly lowers memory usage and computational complexity. This, in turn, allows for faster inference, reduced power consumption, and more efficient utilisation of hardware resources. In extreme cases such as binarized neural networks (BNNs), arithmetic operations can be simplified to bitwise operations, making them especially well-suited for FPGA implementations.

Much of the recent progress in the AI hardware acceleration space can be attributed to the combination of increased hardware specialisation and aggressive quantisation strategies. Specialised architectures are able to exploit the reduced precision of quantised models to maximise parallelism and throughput, achieving performance levels that would be difficult to attain with conventional high precision designs.

This project explores the role of quantisation in FPGA-accelerated neural networks, with a particular focus on binarized neural networks. Using the FINN framework, quantized models are mapped to dataflow-style hardware architectures on an FPGA platform. The study investigates how different levels of quantisation impact performance, as well as how these effects interact with architectural parallelism. Through these experiments, the project aims to provide insight into the trade-offs between accuracy, resource utilisation, and performance in quantised neural network deployment on reconfigurable hardware.

2 System Setup

To evaluate neural network acceleration on FPGA hardware, the FINN framework was deployed on a Diligent Arty-Z7-20 development board. FINN is an open-source framework developed by Xilinx/AMD that enables deployment of quantized neural networkks (QNNs) on FPGA platforms through automated hardware generation and compilation.

2.1 Hardware Platform

The target platform used in this project is the Arty-Z7-20 development board, this board includes an Artix-7 FPGA as the programmable logic (PL) and an ARM Cortex-A9 processor in the form of a Zynq-7000 SOC as the programmable system (PL).

2.2 Software Environment

- PYNQ - A pre-built Linux image for Zynq SOCs (which interfaces with PL using Python) was flashed onto a microSD card and inserted onto the board.
  - There are a small number of pre-built images which AMD maintains, the Arty-Z7-20 isn't one of these boards. Luckily, one of the supported boards (Pynq-Z1), happens to have the exact same specs and (almost the same) pinouts as the Arty-Z7-20, we use the PYNQ v3.0.1 image in this work
- FINN - A compiler which maps Quantized Neural Networks (QNNs) trained in Brevitas to dataflow-style FPGA architectures [1].
- Xilinx Design Suite 2022.2

Figure 1: Image of the Arty-Z7-20

3 Experiment 1: Analysis of the PyTorch to FPGA transformation pipeline for Quantized Neural Networks

4 Experiment 2: Impact of quantisation on neural network FPGA performance

5 Experiment 3: Investigating parallelism in FPGA QNN performance

Basically see if parallelism with certain quantisation sizes has some impact.

References

[1] Yaman Umuroglu, Nicholas J. Fraser, Giulio Gambardella, Michaela Blott, Philip Leong, Magnus Jahre, and Kees Vissers. FINN: A Framework for Fast, Scalable Binarized Neural Network Inference. In *Proceedings of the 2017 ACM/SIGDA International Symposium on Field-Programmable Gate Arrays*, pages 65–74, February 2017. `arXiv:1612.07119`, `doi:10.1145/3020078.3021744`.